

Web Development

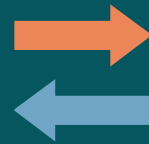
>> Back-End Development

PHP : Intro

What is Web Development?



CLIENT-SIDE
SCRIPTING



DATA



SERVER-SIDE
SCRIPTING



DATABASE
TECHNOLOGY

Server-Side Development

- Websites need to be hosted on a web server
- Server-side code
 - Facilitates the transfer of data from/to web server to a browser
 - Used to build a database or manage data on the web server itself
- Languages
 - There are a number of languages that power the back-end of the web systems

Server-Side Languages



- Powers 75% of web-servers
- Open-source
- Large community
- Cross-platform
- Websites with lower traffic demands



- General purpose, high-level programming language
- Do more with fewer lines of code
- Large standard library
- Like java, used for large amount of traffic



- large-scale websites with a high volume of traffic
- Outperform others raw speed benchmark tests



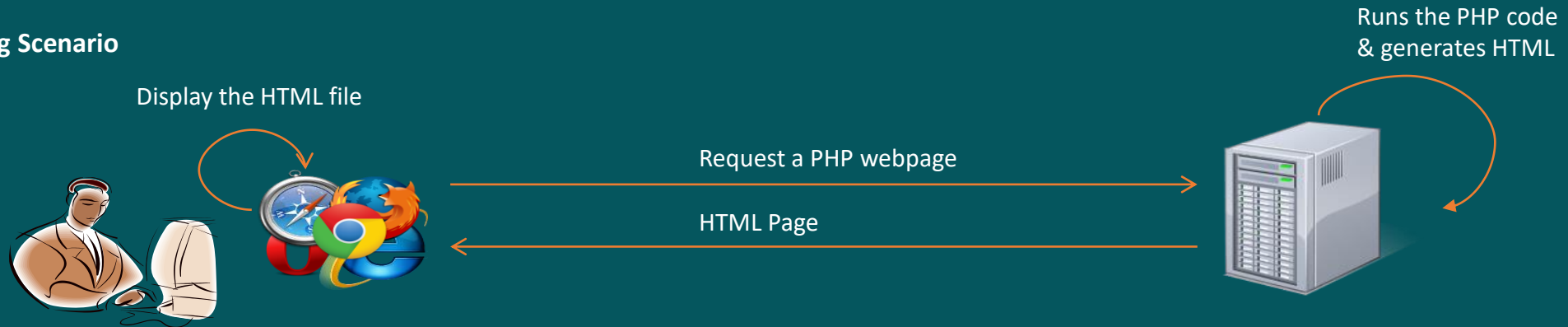
RUBY

- Code is elegant, expressive and powerful
- Speed of development
- Provides a mature framework – Ruby on Rails

PHP- Introduction

- **PHP**
 - **PHP stands for Hypertext Pre-Processor.**
 - Server-Side *Scripting Language*
 - PHP is executed on the server and the result is sent to the browser in plain HTML.
 - Makes Webpages more functional
 - **PHP is an Interpreted language; hence it doesn't need a compiler.**
 - PHP is open-source and free.
 - Cross-platform

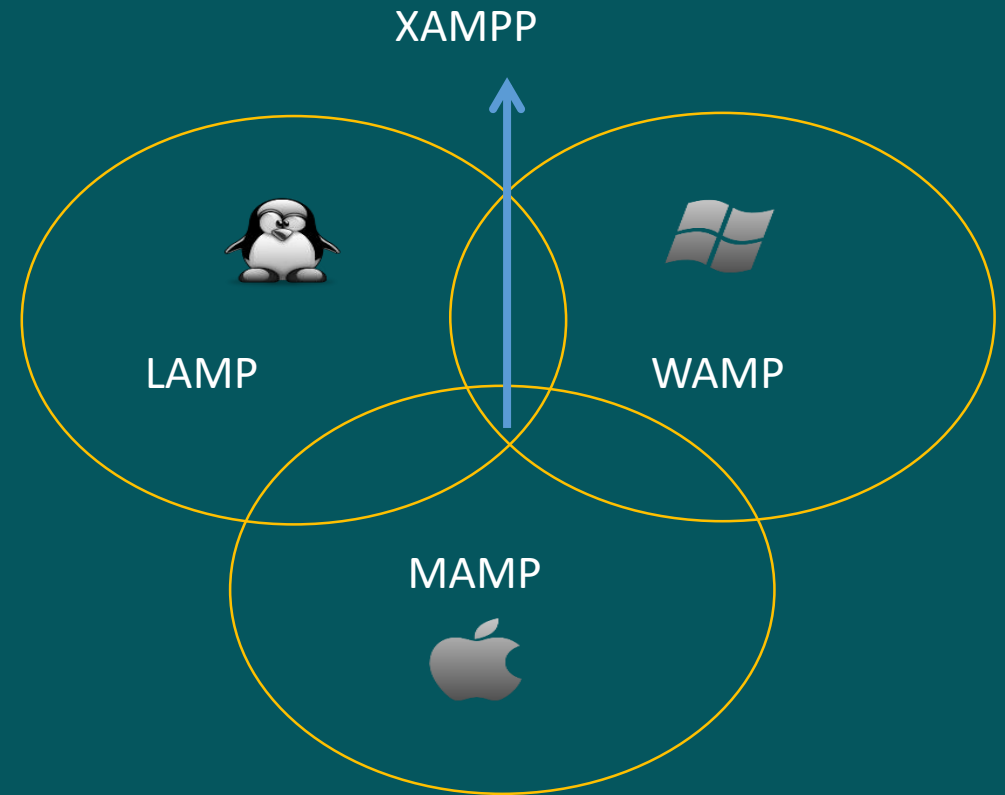
- **Processing Scenario**



Pre-requisite: We need a server that can process and execute PHP

Web Server Solution Package

- Variety of packages available
 - Usually post-fixed as **AMP**
 - Apache
 - MySQL
 - PHP/Perl/Python
 - Package depends on the OS you are using
 - Cross-platforms are also available



Note:

PHP has in-built support for MySQL, which is one of the most widely used Database management systems.

Root Folder

- Root is the place that the webserver will start looking for files when people visit your website
- **Default:** Located for wamp is → C:\wamp\www
- Default: Location for Xamp is → C:\xampp\htdocs
- Set in the **httpd.conf** file (*Apache Configuration File*)

PHP Code Enclosure

- PHP Code is always written between the following pair of delimiters

`<?php statement1; ?>` → the most common

or

`<? statement1; ?>`

or

`<script language = "php"> statement1; </script>`

- **Note:**
 - Always end each PHP statement with a semi-colon
 - Any file containing PHP must have a ".php" file extension

PHP Code Enclosure

```
<?php /* Code Can Go Here */ ?>
<html>
<head>
    <?php /* Code Can Go Here */ ?>
</head>
<body>
    <?php /* Code Can Go Here */ ?>
</body>
</html>
<?php /* Code Can Go Here */ ?>
```

PHP- Create a RandomNumber

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="utf-8">
    <title>Random Number</title>
  </head>
  <body>
    <p>Generating a random number between 1 and 10:
      <?php
        echo rand(1, 10);
      ?>
    </p>
  </body>
</html>
```

Output Statement: Echo

- The “echo” statement is used in PHP to output statements in PHP
 - Similar to the “document.write()” function in JS

```
echo 'New Era to become';
```

```
print "<b>Welcome to my home page <b /> ";
```

Output Statement: Echo

```
<?php  
echo " Today is " . date("l") . ". ";  
?>  
Here's the latest news.
```

Or

```
Today is <?php echo date("l"); ?> Here's the latest  
news.
```

final
output
of the
two
parts

Today is Saturday. Here's the latest news.

Be aware of quotes

- The following examples will cause errors:

```
<?php echo "<p style='color: black;*></p>";  
echo 'What is color?'; ?>
```

```
<?php echo '<p style="color: black;*></p>';
```

```
echo "<p style=\"color: black;\"*></p>";
```

```
echo "What is color?"; echo 'What\'s color?'; ?>
```

Variables

- Variable
 - An entity that can hold a value
- In PHP
 - Variable Names start with “\$” sign
 - Assignment operator is the same “=”

`$name = value;`

NAMING RULES

- Must begin with a letter or an underscore (`_`)
- Must contain only alpha-numeric values and underscore
- No Spaces
- Case-sensitive

Strings

- Strings
 - Like JavaScript, PHP also uses strings
- PHP
 - String concatenation is done using “.” (unlike JavaScript where we use the “+” operator)

```
echo $txt1 . " " . $txt2;
```

- Provides a number of string processing functions

<http://devdocs.io/php-string/>

What is the output?

```
<?php
```

```
    $x = 10;
```

```
    $y = 20;
```

```
    $z = $x+$y;
```

```
    echo "(before) z = ". $z . "\n";
```

```
    $y=5;
```

```
    echo "(after) z = ". $z . "";
```

```
?>
```


What is the output

```
<?php
```

```
$x = 4;
```

```
function assignx () {
```

```
    $x = 0;
```

```
    echo "\$x inside function is $x. \n";
```

```
}
```

```
    assignx();           //function call
```

```
    echo "\$x outside of function is $x. \n";
```

```
?>
```

What is the output

```
<?php
```

```
$x = 4;
```

```
function assignx () {
```

```
    echo "\$x inside function is $x. \n";
```

```
}
```

```
assignx();    //function call
```

```
?>
```

What is the output

```
<?php
```

```
$x = 4;
```

```
function assignx () {
```

```
    GLOBAL $x ;
```

```
    echo "\$x inside function is $x. \n";
```

```
}
```

```
assignx();          /* function call */
```

```
?>
```

A constant

- It is typically defined near the top of a PHP file via the `define()` function,

```
<?php
# uppercase for constants is a programming convention
define("DATABASE_LOCAL", "localhost");
define("DATABASE_NAME", "ArtStore");
define("DATABASE_USER", "Fred");
define("DATABASE_PASSWD", "F5^7%ad");
...
# notice that no $ prefaces constant names
$db = new mysqli(DATABASE_LOCAL, DATABASE_USER,
DATABASE_PASSWD ,
DATABASE_NAME);
```

Operators

Arithmetic

Operator	Name
$x + y$	Addition
$x - y$	Subtraction
$x * y$	Multiplication
x / y	Division
$x \% y$	Modulus

Inc / Dec

Operator	Name
$++ x$	Pre-increment
$x ++$	Post-increment
$-- x$	Pre-decrement
$x --$	Post-decrement

Logical

Operator	Name
$x \text{ and } y$	And
$x \text{ or } y$	Or
$x \text{ xor } y$	Xor
$x \&\& y$	And
$x y$	Or
$! x$	Not

Comparison

Operator	Name
$x == y$	Equal
$x === y$	Identical
$x != y$	Not equal
$x <> y$	Not equal
$x !== y$	Not identical
$x > y$	Greater than
$x < y$	Less than
$x >= y$	Greater than or equal to
$x <= y$	Less than or equal to

Conditional

- Conditionals
 - Provides a way to have certain commands executed only if some condition is met
- PHP
 - If statement
 - If.....Else statement
 - Switch statement

Conditional Statements

Switch statement

SYNTAX

```
switch (n)
{
    case label1:
        code to be executed if n=label1;
        break;
    .....
    default:
}
```

If – else statement

SYNTAX

```
if(cond)
{

}
else
{

}
```

If statement

SYNTAX

```
if(cond)
{

}
```

If-Else statement with Complex Conditional

SYNTAX

```
if(cond)
{

}
else if
{

}
```

Example

```
<?php  
$x=1;  
if ($x == 1)  
    echo '$x is equal to 1';  
?>
```


Example

```
<?php
$x=1;
if ($x == 1)
{
    echo '$x is equal to 1';
    $x++;
    echo 'now $x is equal to 2';
}
?>
```

Example

```
<?php
    $x=1;
    if ($x == 1 OR $x == 2)
        echo '$x is equal to 1 (or maybe 2) ' ;
?>
```

Example

```
<?php
    $x=1;
    if ($x == 1 AND $x == 2)
        echo '$x is equal to 1 (or maybe 2) ' ;
?>
```

Else Statements

```
<?php
    $x=1;
if ($x == 2) {
    echo '$x is equal to 2'; }
else
    { echo '$x is equal to 1'; }
?>
```

Example

```
<?php
    $x=1;
    if ($x == 1 AND $x == 2)
        echo '$x is equal to 1 (or maybe 2) ' ;
else
    echo 'one of the conditions is not true (or
both) ' ;
?>
```

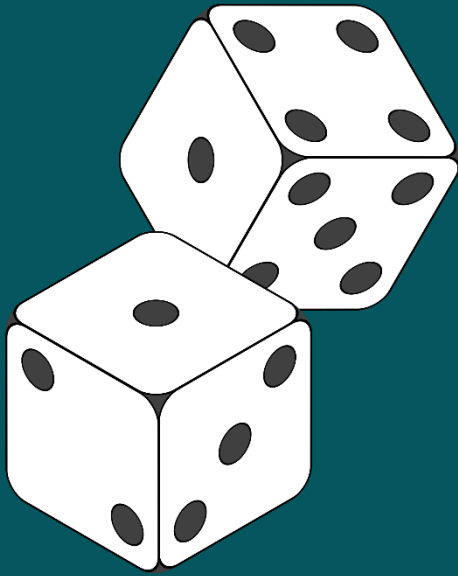
Else Statements

```
<?php
    $x=1;
if ($x == 2) {
    echo '$x is equal to 2'; }
else if ($x == 1)
    { echo '$x is equal to 1'; }
else { echo '$x does not equal 2 or 1'; }

?>
```

Example

PHP-DiceRoll-TwoDice



```
$roll1 = rand(1, 6);  
$roll2 = rand(1, 6);  
echo '<p>You rolled a ' . $roll1 . ' and a ' . $roll2  
. '</p>';  
    if ($roll1 == 6 && $roll2 == 6) { echo  
'<p>You win!</p>';  
    }  
    else {  
  
echo '<p>Sorry, you didn't win, better luck next  
time!</p>';  
    }  
echo '<p>Thanks for playing</p>';
```

Loops

- Looping
 - Provides a way to have certain block of commands to be executed again and again for certain number of times
- PHP
 - While Loop
 - Do....While Loop
 - For Loop
 - ForEach

Loop Statements

While statement

SYNTAX

```
while(cond)
{
}
```

do-while statement

SYNTAX

```
do
{
}
while(cond);
```

At least runs the loop once

for statement

SYNTAX

```
for(init; cond; step)
{
}
```

Example

```
<html>
  <body>

    <?php
      $c = 0;
      $d = 0;

      for( $i = 0; $i<5; $i++ ) {
        $c += 10;
        $d += 5;
      }

      echo ("At the end of the loop a =
$c and d = $d");
    ?>

  </body>
</html>
```

At the end of the loop a = 50 and d = 25

Example

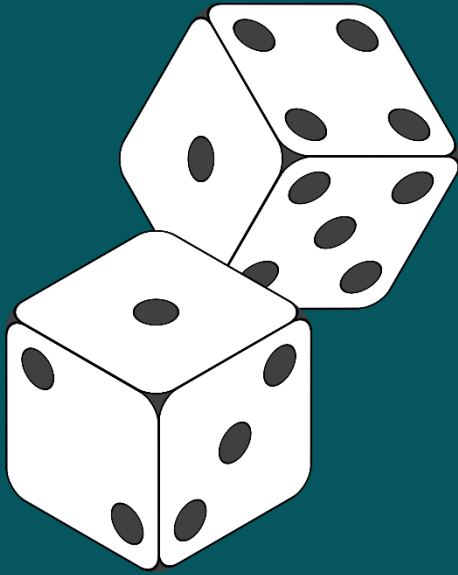
```
<?php
for ($count = 1; $count <= 10;$count = $count + 3)
{
echo $count . ' ';
}
?>
```

This will result in the following:

1 4 7 10

Example

PHP-DiceRoll-SingleDice



```
for ($count = 1; $count <= 10; $count++) {  
  
    $roll = rand(1, 6);  
  
    echo '<p>You rolled a ' . $roll . '</p>';  
    if ($roll == 6) {  
  
        echo '<p>You win!</p>';  
  
    }  
  
    else {  
  
        echo '<p>Sorry, you didn't win, better luck  
next time!</p>';  
  
    }  
  
    }  
  
    echo '<p>Thanks for playing</p>';
```

While Loop

```
<html>
  <body>
    <?php
      $i = 0;
      $num = 50;

      while ($i < 10) {
        $num--;
        $i++;
      }
      echo ("Loop stopped at i = $i and
num = $num");
    ?>

  </body>
</html>
```

Loop stopped at $i = 10$ and $num = 40$

While Loop

```
<html>
  <body>
    <?php
      $i = 10;
      $num = 50;

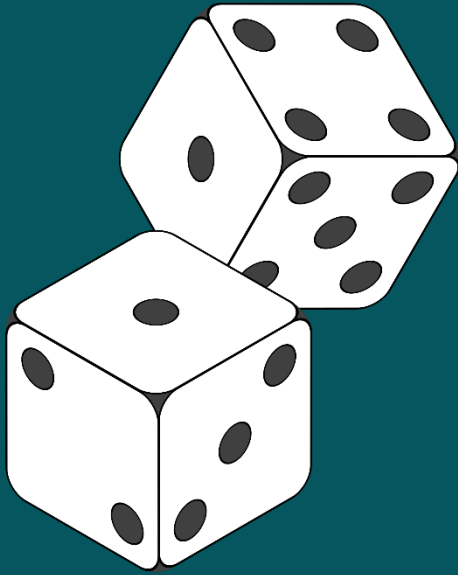
      while ($i < 10) {
        $num--;
        $i++;
      }
      echo ("Loop stopped at i = $i and
num = $num");
    ?>

  </body>
</html>
```

Loop stopped at i = 10 and num = 50

Example

PHP-DiceRoll-SingleDice



```
$roll = 0;
```

```
while ($roll != 6) {
```

```
    $roll = rand(1, 6);
```

```
    echo '<p>You rolled a ' . $roll . '</p>';
```

```
    if ($roll == 6) {
```

```
        echo '<p>You win!</p>';
```

```
    }
```

```
    else {
```

```
        echo '<p>Sorry, you didn\'t win, better luck next  
time!</p>';
```

```
    }
```

```
}
```

Do While Loop

```
<html>
  <body>

    <?php
      $i = 0;
      $num = 0;

      do {
        $i++;
      }

      while ($i < 10);
      echo ("Loop stopped at i = $i");
    ?>

  </body>
</html>
```

Loop stopped at i = 10

Do While Loop

Do While is executed at least **Once**.

```
<html>
  <body>

    <?php
      $i = 10;
      $num = 0;

      do {
        $i++;
      }

      while ($i < 10);
      echo ("Loop stopped at i = $i");
    ?>

  </body>
</html>
```

Loop stopped at i = **11**

Thanks